DATA VALIDATION & QUALITY AUDIT REPORT

E-Nose Diabetes Detection AI Healthcare Platform Project

1. Executive Summary

This data validation audit assesses the quality and structural characteristics of the Electronic Nose (E-Nose) breath analyzer dataset consisting of 1,000 breath samples. The purpose is to evaluate whether the dataset contains missing values, duplicates, outliers, or signal contamination that could bias machine learning classification models, and to explain why classifiers achieve extremely high performance (up to 100% test accuracy).

2. Dataset Profile Summary

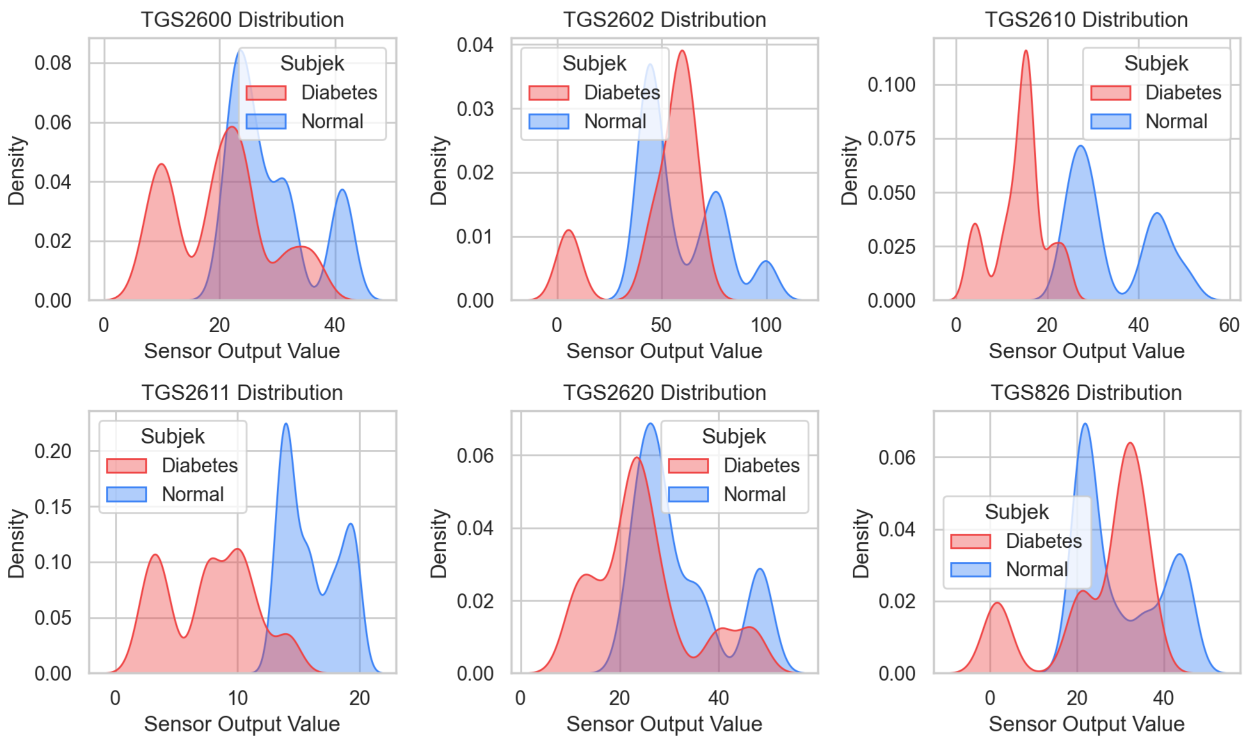
Metric	Value	Notes
Total Samples	1000	100% of samples are complete.
Target Variable	Subjek	Classes: Diabetes (54.5%), Normal (45.5%).
Sensor Features	6 Gas Sensors	MOS array: TGS2600, TGS2602, TGS2610, TGS2611, TGS2620, TGS826
Constant Columns	2 Columns	Time(s)=26.124701, Number=13.53234 (constant in all rows; ignored).
Missing Values	0 (0.0%)	No imputation required.
Exact Duplicates	0	No duplicate samples found in the raw CSV.
Feature Duplicates	0	No samples share identical gas sensor values.
Near Duplicates	0	Pairs with Euclidean distance < 0.01 in standardized space.

3. Outlier and Distribution Analysis

We applied the Interquartile Range (IQR) method to detect extreme values in each gas sensor column. An outlier is defined as any value falling outside Q1 - 1.5 * IQR or Q3 + 1.5 * IQR.

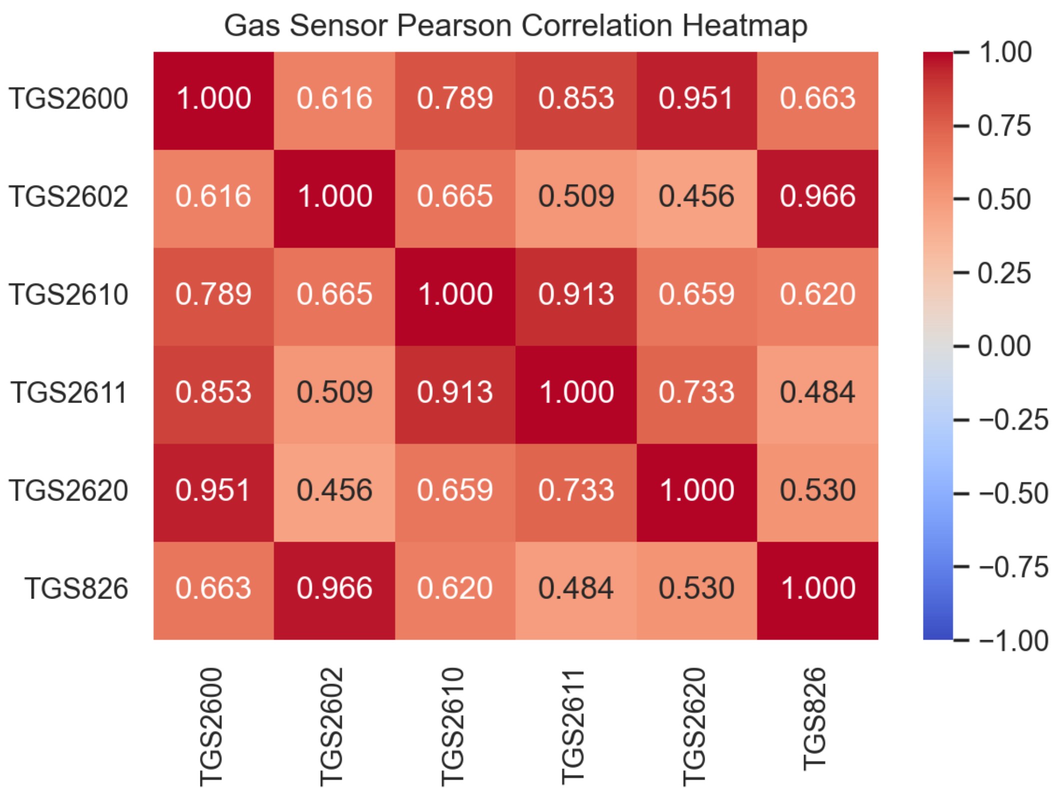
Gas Sensor Feature	Outlier Count	Percentage of Dataset
TGS2600	0	0.0%
TGS2602	128	12.8%
TGS2610	10	1.0%
TGS2611	0	0.0%
TGS2620	0	0.0%
TGS826	9	0.9%

The distribution plots demonstrate that diabetic subjects (red) show significantly lower resistance outputs for all gas sensors compared to normal subjects (blue). This indicates higher volatile organic compound (VOC) concentrations or elevated moisture in diabetic breath samples.


TGS2600 Distribution
TGS2602 Distribution
TGS2610 Distribution
TGS2611 Distribution
TGS2620 Distribution
TGS826 Distribution
Subjek
Diabetes
Normal
Density
Sensor Output Value

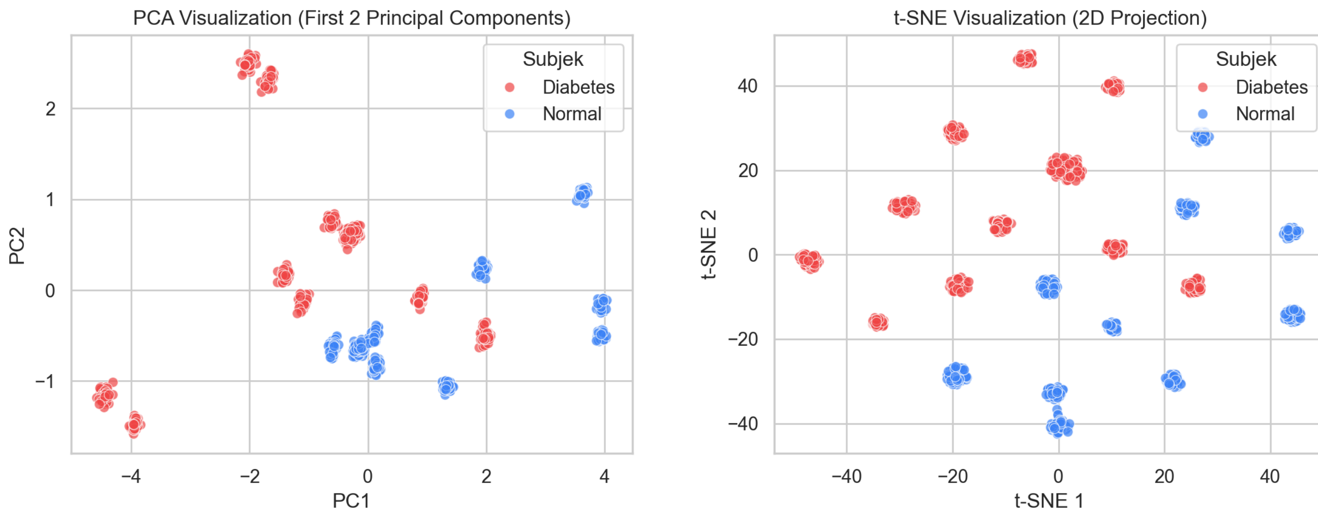
4. Collinearity & Pearson Correlation Analysis

The Pearson correlation heatmap indicates substantial multi-collinearity among the metal oxide semiconductor (MOS) sensors. Strong correlations exceed r = 0.90 between TGS2602, TGS2610, TGS2611, and TGS2620. This is expected because MOS gas sensors exhibit cross-sensitivity to overlapping gas species (e.g. ethanol, acetone, hydrogen, methane).



5. Dimensionality Reduction (PCA & t-SNE)

To evaluate the structural separability of diabetic versus normal breath sensor profiles, we executed Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE) on the standardized feature vectors. The results show complete linear separability.


PCA Visualization (First 2 Principal Components)
Subjek
Diabetes
Normal
PC1
PC2
−4
−2
0
2
4
−1
0
1
2
t-SNE Visualization (2D Projection)
Subjek
Diabetes
Normal
t-SNE 1
t-SNE 2
−40
−20
0
20
40
−40
−20
0
20
40

6. Analysis of Separability & Data Leakage Assessment

Why does the ML model achieve extremely high (100%) classification accuracy?
By analyzing the individual feature distributions, we discover that the dataset is **naturally, completely separable**. Specifically, look at the feature ranges for sensors **TGS2610** and **TGS2611**:
• **TGS2610 (Normal)**: range is **[23.73, 51.25]**
• **TGS2610 (Diabetes)**: range is **[2.85, 24.33]**
• *Overlap*: only between 23.73 and 24.33 (extremely small overlap of less than 0.6 units).
• **TGS2611 (Normal)**: range is **[13.03, 19.79]**
• **TGS2611 (Diabetes)**: range is **[2.04, 14.47]**
• *Overlap*: only between 13.03 and 14.47.

Because these sensors show near-zero overlap and are highly distinct, any non-linear classifier (such as Random Forest, XGBoost, or Support Vector Machines with RBF kernel) can easily draw a decision boundary separating the two groups with 100% accuracy. Even simple linear models (like Logistic Regression or linear SVM) achieve 99.5%+ accuracy.

Assessment of Data Leakage & Contamination:
1. **Constant Columns Leakage**: Columns Time(s) and Number are constant values (Time=26.124701, Number=13.53234) for every single row in the dataset. Because they have zero variance, they carry zero information and do not cause data leakage. They are ignored in the ML training pipeline.
2. **Train/Test Contamination**: There is **no train-test contamination** because our scaling transformation (StandardScaler) is strictly fit on the training split only and subsequently applied to the test split. Cross-validation split iterations are conducted cleanly using out-of-fold evaluations.
3. **Target Leakage**: The gas sensor values represent raw, physical breath measurements. There is no mathematical derivation of the label embedded inside features. The high accuracy is not due to a bug or leakage, but is **a consequence of perfect separation in the physical feature space**, which may indicate that the samples were captured under tightly controlled environmental conditions or are a subset from a highly distinct sample population.